

EDUCATION

Department of Computer Science, University of Victoria

Victoria, Canada

Ph.D. in Computer Science

2024 - Present

- Research focus: *Performance Analysis and Optimization of Networked Systems*

School of Cyber Science and Engineering, Southeast University

Nanjing, China

M.E. in Computer Network Engineering

2021 - 2024

- Recommended Graduate Student [\[Transcript\]](#)

- Research focus: *Network Traffic Analysis & High-Speed Data Stream Processing*

School of Cyber Science and Engineering, Southeast University

Nanjing, China

B.E. in Computer Network Engineering

2017 - 2021

- GPA:3.63/4.0 [\[Transcript\]](#)

PUBLICATIONS

- [1] **Quanwei Zhang**, Zhiming Huang, Jinwei Zhao, Ali Ahangarpour, Jianping Pan. “Faster Exploration and Exploitation for Communication Environment Awareness in Starlink” *IEEE INFOCOM*, 2026.
- [2] **Quanwei Zhang**, Jianping Pan. “Decentralized Gossip Learning with Adaptive OOD Detection for LEO Satellite Networks under Non-IID Data” *IEEE GLOBECOM*, 2025.
- [3] **Quanwei Zhang**, Zhiming Huang, Jinwei Zhao, Jianping Pan. “A Congestion Control Test Suite for Real-Time Communication” *ACM MMSys*, 2025.
- [4] Jiachen Yang, Jiankun Peng, **Quanwei Zhang**, et al. “Monocular Vision Approach for Soft Actor-Critic Based Car-Following Strategy in Adaptive Cruise Control” *Expert Systems with Applications*, 2025. [\[Link\]](#)
- [5] **Quanwei Zhang**, Qingjun Xiao, Yuexiao Cai. “A Generic Sketch for Estimating Super-Spreaders and Per-Flow Cardinality Distribution in High-Speed Data Streams” *Computer Networks*, 2023. [\[Link\]](#)
- [6] Qingjun Xiao, Lin Wen, **Quanwei Zhang**. “Multi-Resolution Odd Sketch for Mining Jaccard Similarities between Dynamic Streaming Sets” *IEEE CSCWD*, 2021. [\[Link\]](#)
- [7] **Quanwei Zhang**, Dazhong Li, Yue Fei, et al. “RDCPF: A Redundancy-Based Duty-Cycling Pipelined-Forwarding MAC for Linear Sensor Networks” *Sensors*, 2020. [\[Link\]](#)

PATENTS

- [1] Qingjun Xiao, **Quanwei Zhang**, Yuexiao Cai. “Method for Estimating High-Speed Network Progressive Base and Distribution Function Based on General Outline”. China Patent CN117336209B, Granted. [\[Link\]](#)
- [2] Fei Tong, **Quanwei Zhang**, Dazhong Li, et al. “Linear Sensor Network Multi-Hop Data Collection Method Based on Redundant Nodes” China Patent CN112073931B, Granted. [\[Link\]](#)

CONFERENCE PRESENTATIONS

- “Faster Exploration and Exploitation for Communication Environment Awareness in Starlink”, oral presentation at IEEE INFOCOM’26 May 2026
- “A Congestion Control Test Suite for Real-Time Communication”, poster pitch at ACM MMSys’25 (remote) April 2025

TEACHING EXPERIENCE

- Tutorial Instructor and marker for CSC360: Operating Systems.

Fall 2025 & Spring 2026

PROFESSIONAL ACTIVITY

- Reviewer, IEEE/ACM Transactions on Networking
- Web Administrator (Webmaster), IEEE Victoria Section
- LinkedIn Page Administrator, IEEE Women in Engineering (WiE) Victoria Section

AWARDS AND HONORS

- IEEE INFOCOM Student Travel Grant 2026
- University of Victoria Fellowship 2024 - 2025
- National Scholarship, China 2023
- Graduate Academic Scholarship 2021 - 2024